

SOP-API-01: Standard Operating Procedure - API Deployment Validation

Details

Category: SOP | Service: Payment API, Order Service, User Service | Department: Platform Engineering
| Version: 3.0

Section 1 - Pre-deployment checklist

Verify the release manifest references an existing ConfigMap and Kubernetes Secret for the target namespace. Confirm environment variables match the vault entries. Validate database migration scripts on a staging replica before promotion.

Section 2 - Secret and configuration validation

Inspect Kubernetes Secret values with `kubectl get secret payment-secrets -o yaml` and compare checksums with the vault. An incorrect Kubernetes Secret or a missing Secret key is a frequent cause of HTTP 503 and CrashLoopBackOff after deployment. Re-apply the corrected Secret with `kubectl apply` and restart affected pods.

Section 3 - Post-deployment monitoring

Inspect application logs for the first 15 minutes after rollout. Monitor the 5xx error rate dashboard, pod restart counters and JVM heap usage. If the error rate exceeds 1 percent, halt the rollout immediately.

Section 4 - Health probe requirements

Every service must expose `/healthz` (liveness) and `/readyz` (readiness). Liveness probe timeout must be at least 5 seconds to avoid restart loops on cold start. Readiness probes must fail while database connectivity is unavailable so the load balancer removes the pod from rotation.